

Apple & Crabapple Disease Management

Covers: Apple Scab, Cedar-Apple Rust, Black Rot (Frogeye Leaf Spot), Fire Blight, Powdery Mildew

Compiled from public university cooperative extension publications for testing InsightAI-RAG's retrieval and grounding quality against real agricultural guidance. This is not a substitute for consulting a local agronomist or extension office before applying any treatment. Always follow current product labels; pesticide registrations change and vary by region.

Apple Scab — *Venturia inaequalis*

Apple scab begins in spring as dark, olive-green leaf spots less than half an inch in diameter. Severe infections can affect the entire leaf, causing it to turn brown and drop from the tree. Slightly raised, black spots deform the fruit.

Prevention & Treatment: Plant resistant varieties for best control ('Goldrush', 'Enterprise', 'Pristine', 'Gala Supreme', 'Liberty', 'Freedom', 'Jonafree' are scab-resistant). Rake and remove fallen leaves to reduce early spring infection sources. If chemical control is warranted, use captan or sulfur fungicide, applied according to label directions.

Source: Clemson Cooperative Extension, HGIC 2000 'Apple & Crabapple Diseases' (updated Jul 2025)

Cedar-Apple Rust — *Gymnosporangium juniperi-virginianae*

This fungal disease requires Eastern red cedar or other ornamental junipers to complete its life cycle, alternating between the two hosts. Bright orange-yellow raised spots appear on apple/crabapple leaves in mid- to late spring; severe defoliation can reduce bloom the following season. Infected fruit is often small and distorted. Hard brown galls form on cedar branches and, after spring rain, produce gelatinous orange spore tendrils that can blow up to half a mile to infect nearby apple trees.

Prevention & Treatment: Plant resistant varieties. Remove nearby Eastern red cedars or prune out galls where possible. If chemical control is warranted, use myclobutanil or sulfur fungicide on apples.

Source: Clemson Cooperative Extension, HGIC 2000 'Apple & Crabapple Diseases' (updated Jul 2025); University of Kentucky Plant Pathology Fact Sheet PPFS-FR-T-05

Black Rot (Frogeye Leaf Spot) — *Botryosphaeria obtusa* (*Phyalospora obtusa*)

Begins as a purple leaf speck that enlarges into a brown or tan center resembling a frog's eye; heavily infected leaves drop, weakening the tree and reducing next year's flowering. Fruit infections start as tiny red/purple spots opposite the stem end, enlarging into alternating black and brown zones; affected fruit wrinkles, mummifies, and often stays attached to the tree. Limb cankers appear as slightly sunken, reddish-brown areas.

Prevention & Treatment: Remove and discard dead branches and mummified fruit, where the fungus overwinters. Captan fungicide is effective if applied early and at regular intervals through the season.

Source: Clemson Cooperative Extension, HGIC 2000 'Apple & Crabapple Diseases' (updated Jul 2025)

Fire Blight — *Erwinia amylovora*

A fast-moving bacterial disease that develops rapidly in early spring during rainy weather above 60F while the tree is in bloom. Blossoms and young twigs wilt, shrivel, and turn brown to black; twig tips die and curl into a characteristic 'shepherd's crook.' A milky, sticky bacterial ooze can appear on stems in wet weather.

Prevention & Treatment: Remove blighted twigs and cankers before spring growth starts, pruning 12-18 inches below any sign of infected tissue and disinfecting tools between cuts (1 part bleach to 9 parts water). Avoid high-nitrogen fertilization, which increases susceptible succulent growth. Where chemical control is warranted, streptomycin applied during bloom (3 applications at 3-day intervals, starting at bloom onset, above 60F) is used commercially; a copper fungicide during dormancy is an option if fire blight is expected to be severe.

Source: Clemson Cooperative Extension, HGIC 2000 'Apple & Crabapple Diseases' (updated Jul 2025)

Healthy

No signs of disease detected. Maintain regular watering, balanced fertilization, and routine pest/disease scouting to keep trees productive.

Source: General extension guidance